Bacterial de novo Biosynthesis of Thiamine Diphosphate (ThDP)

A review-style synthesis for a molecular-biology audience Search date: 2026-07-15. Literature accessed via PubMed.

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1. Executive summary

Thiamine diphosphate (ThDP, also thiamine pyrophosphate/TPP) is the catalytically active form of vitamin B1 and an obligatory cofactor for enzymes that break or form C–C bonds adjacent to a carbonyl (pyruvate dehydrogenase, 2-oxoglutarate dehydrogenase, transketolase, pyruvate decarboxylase, DXP synthase). Most bacteria synthesize ThDP de novo, whereas humans cannot — making the pathway a long-standing reservoir of antibacterial and herbicide targets.

The bacterial pathway is best understood as **two independently produced ring systems that are welded together and then activated**:

1. A **hydroxymethylpyrimidine (HMP) branch**, in which the radical-SAM enzyme **ThiC** rearranges the purine-pathway intermediate AIR into HMP-phosphate, and the kinase **ThiD** raises it to HMP-pyrophosphate (HMP-PP).
2. A **thiazole branch**, in which **Dxs** supplies the five-carbon skeleton (DXP), a glycine/tyrosine-derived enzyme supplies **dehydroglycine**, a protein-borne **sulfur relay (IscS → ThiI/ThiF → ThiS-thiocarboxylate)** supplies sulfur, and **ThiG** condenses all three into thiazole phosphate.

ThiE (thiamine-phosphate synthase) then couples HMP-PP + thiazole-P into thiamine monophosphate (TMP), and **ThiL** (thiamine-monophosphate kinase) phosphorylates TMP to ThDP. The chemistry is remarkable for two of the most unusual reactions in cofactor biology (ThiC's radical rearrangement and ThiG's convergent ring assembly) and for the use of an **ubiquitin-like sulfur-carrier protein (ThiS)** rather than free sulfide.

The pathway's most important **variation** is in how sulfur and dehydroglycine are supplied: aerobes such as *Bacillus subtilis* use the flavoenzyme **ThiO** (glycine oxidase), while facultative anaerobes such as *Escherichia coli* use the oxygen-sensitive radical-SAM enzyme **ThiH** (tyrosine lyase). ThiI's domain architecture and even its necessity differ between Gammaproteobacteria and Firmicutes, and redundant sulfur routes exist. The **final activation step also diverges from eukaryotes**: bacteria make ThDP in one ThiL step from TMP, whereas eukaryotes hydrolyze TMP to free thiamine and pyrophosphorylate it (TPK/ThiN).

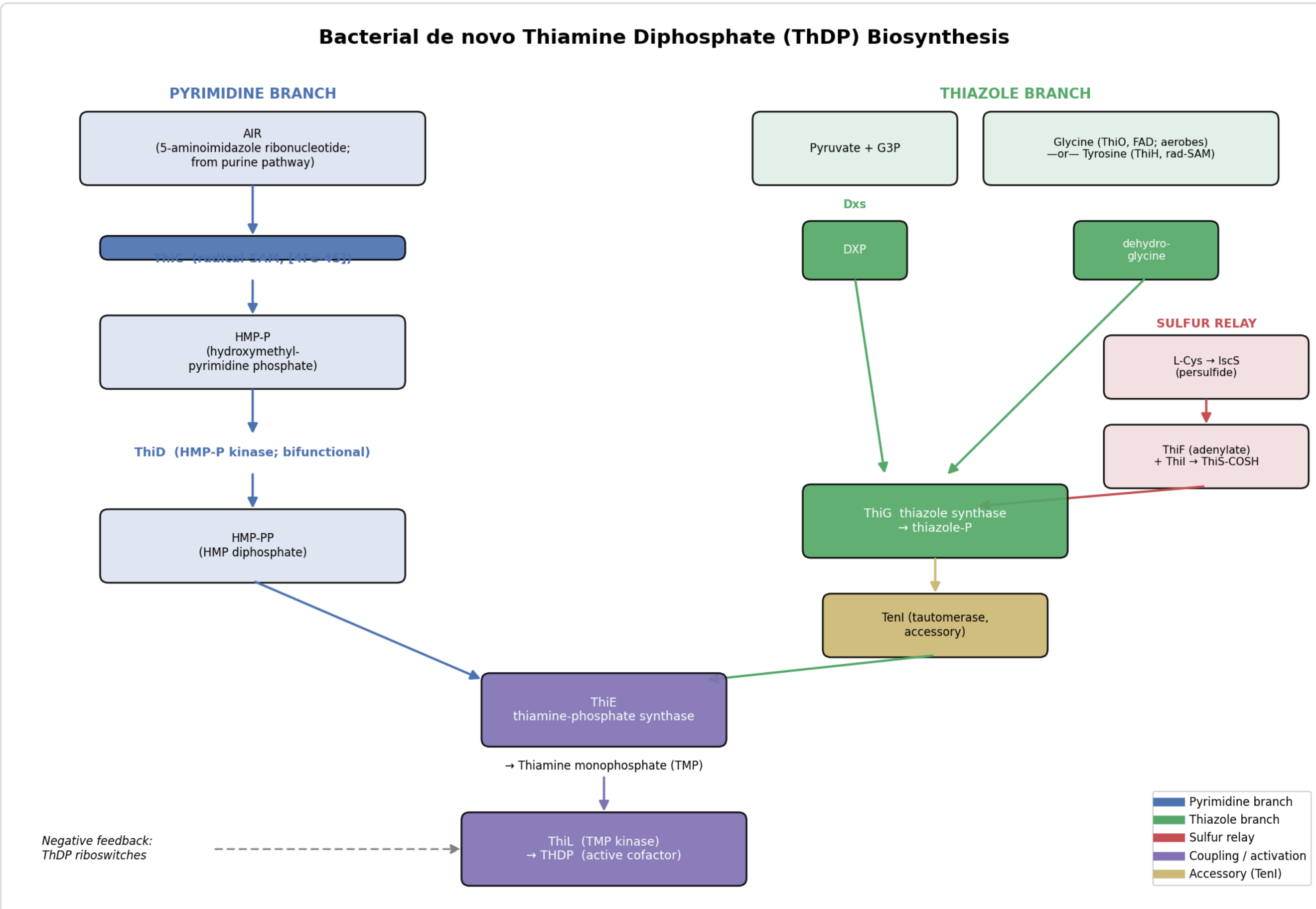

Figure 1. *Schematic of the bacterial de novo ThDP biosynthesis pathway showing the pyrimidine branch (AIR → ThiC → HMP-P → ThiD → HMP-PP), the thiazole branch (Dxs/DXP + ThiO/ThiH-derived dehydroglycine + IscS/ThiF/ThiI/ThiS sulfur relay → ThiG → thiazole-P, with accessory TenI tautomerase), their convergence at ThiE (→ TMP), and final activation by ThiL (→ ThDP), under ThDP-riboswitch feedback.*

Figure 1. Overview of bacterial de novo ThDP biosynthesis. Two independently synthesized rings (pyrimidine, blue; thiazole, green) are joined by ThiE and activated by ThiL. The thiazole branch integrates three inputs onto ThiG: the DXP carbon skeleton (Dxs), dehydroglycine (ThiO in aerobes or ThiH in anaerobes), and protein-carried sulfur (red relay). TenI (accessory) tautomerizes the thiazole product. ThDP riboswitches provide end-product feedback.

2. Definition and biological boundaries

What is inside the system

The system is the **cytoplasmic, de novo enzymatic route that converts central metabolites into ThDP** in bacteria. Its obligatory members are:

Branch	Step	Enzyme	Reaction
Pyrimidine	HMP-P formation	**ThiC**	AIR → HMP-P (radical SAM)
Pyrimidine	HMP-P activation	**ThiD**	HMP-P → HMP-PP (two kinase steps)
Thiazole	C5 skeleton	**Dxs**	pyruvate + G3P → DXP
Thiazole	glyoxyl imine	**ThiO** *or* **ThiH**	glycine *or* tyrosine → dehydroglycine
Thiazole	sulfur mobilization	**IscS**, **ThiI**, **ThiF**, **ThiS**	Cys → ThiS-thiocarboxylate
Thiazole	ring assembly	**ThiG**	DXP + dehydroglycine + S → thiazole-P
Coupling	condensation	**ThiE**	HMP-PP + thiazole-P → TMP
Activation	final phosphorylation	**ThiL**	TMP → ThDP

Neighboring processes that are often conflated but should be treated separately

- **Thiamine salvage and transport.** Uptake of exogenous thiamine (e.g., YuaJ/PnuT permeases, ABC transporters) and salvage enzymes feed ThDP pools by routes that **bypass** de novo synthesis. Salvage recycles the two rings: **ThiM** (thiazole kinase) rephosphorylates recovered thiazole (THZ) (P 26634770 P 23816351) and **TenA** (thiaminase II) regenerates the aminopyrimidine, while HMP kinase feeds HMP-P. Crucially, salvage and de novo **share ThiD and ThiE**, so these two are common nodes; ThiM and TenA are salvage-specific. In *S. aureus* the salvage genes (TenA–ThiD–ThiM–ThiE) form one operon and TPK another, both under TPP-riboswitch and enzymatic control (P 19888457). In *B. subtilis*, thiamine is not a pathway intermediate but is salvaged and pyrophosphorylated by a separate ThiN (TPK) route (P 16291685). These are adjacent modules, not part of de novo synthesis.

- **The MEP/isoprenoid pathway. Dxs is shared**: DXP is simultaneously the first committed intermediate of the methylerythritol-phosphate (MEP) pathway for isoprenoids and a thiazole precursor. Dxs is therefore a **branch-point enzyme**, not a thiamine-dedicated one (P 41994131 P 40845551). Attributing Dxs solely to thiamine biology is a common oversimplification.
- **The IscS/SUF sulfur-trafficking network.** IscS (cysteine desulfurase) is a hub serving Fe–S cluster assembly, thio-modification of tRNA (4-thiouridine), molybdopterin, biotin, lipoic acid and NAD in addition to thiamine (P 12382038). ThiI likewise is shared between thiamine and 4-thiouridine synthesis. The sulfur machinery **overlaps** the thiamine pathway but is not exclusive to it.
- **Purine biosynthesis.** ThiC consumes AIR (5-aminoimidazole ribonucleotide), a purine-pathway intermediate. The pyrimidine ring of thiamine thus derives from **purine**, not pyrimidine-nucleotide, metabolism — a frequent point of confusion.

Competing definitions

Gene nomenclature is not uniform: *E. coli* thiazole synthase is a **ThiGH** heterodimer, whereas *Bacillus* uses **ThiG** + **ThiO**; ThiD is sometimes annotated as the bifunctional **ThiDIJ** (HMP kinase + HMP-P kinase) (P 10075431). Eukaryotic thiazole synthesis (fungal/plant **THI4**) is mechanistically unrelated and should not be merged with the bacterial ThiG system (see §5).

3. Mechanistic overview (best current model)

Pyrimidine branch. ThiC catalyzes one of the most complex known rearrangements in cofactor biology: the near-complete reorganization of AIR into HMP-P. It is an iron–sulfur, radical-S-adenosylmethionine (SAM) enzyme; a C-terminal CX2CX4C motif ligates a [4Fe-4S] cluster that reductively cleaves SAM to a 5'-deoxyadenosyl radical, abstracting a hydrogen to initiate rearrangement. HMP-P and 5'-deoxyadenosine are the products (P 18953358).

ThiD then performs two sequential phosphorylations — HMP → HMP-P and HMP-P → HMP-PP — and in *E. coli* is a single bifunctional protein (ThiD/J) (P 10075431 P 8885414). Only HMP-PP is competent for coupling.

Thiazole branch (three converging inputs onto ThiG). 1. *Carbon skeleton:* **Dxs** condenses pyruvate and glyceraldehyde-3-phosphate to **DXP** (a ThDP-dependent reaction — the pathway partly bootstraps on its own product). 2. *Glyoxyl imine / dehydroglycine:* generated by **ThiO** (FAD-dependent glycine oxidase; oxidizes glycine to the imine) in aerobes, or by **ThiH** (radical-SAM tyrosine lyase that cleaves tyrosine's Cα–Cβ bond, releasing p-cresol) in *E. coli* (P 12627963 P 19923213). 3. *Sulfur:* **IscS** abstracts sulfane sulfur from L-cysteine as a protein persulfide (P 12382038); **ThiF** adenylates the C-terminal glycine of **ThiS** with ATP; with **ThiI** participation the acyl-adenylate is converted to a **C-terminal thiocarboxylate** on ThiS (P 9632726). ThiS-COSH is the immediate sulfur donor.

ThiG then condenses DXP, dehydroglycine and the ThiS-thiocarboxylate sulfur into **thiazole phosphate**. Isotope labeling shows the ThiS sulfur is delivered into the ring and a thioenolate intermediate is trapped; strikingly, the sulfur that leaves ThiS is replaced by an oxygen derived from DXP

(P 15489164).

The complete E. coli system has been reconstituted: thiazole synthase activity requires the **ThiGH complex, tyrosine, DXP and cysteine plus IscS and ThiI** (six structural proteins in total), and is stimulated by SAM and NADPH — directly confirming the multi-protein, tyrosine-fed assembly in this organism (P 14757766).

In many bacteria an accessory tautomerase, **TenI**, then isomerizes the ThiG product to the thermodynamically favored thiazole tautomer that is the true ThiE substrate; TenI is a ThiE-like (βα)8 fold that has *lost* thiamine-phosphate synthase activity and instead uses phosphate/His122 as acid–base catalysts (P 21534620).

Coupling and activation. **ThiE** (thiamine-phosphate synthase), a (β/α)8-barrel enzyme, condenses HMP-PP and thiazole-P with loss of pyrophosphate to give **thiamine monophosphate (TMP)** (P 22226942).

ThiL phosphorylates TMP to **ThDP** using ATP, the final and committed activation step (P 30867460 P 32404369).

Obligatory vs conditional vs accessory. - *Obligatory:* ThiC, ThiD, Dxs, ThiG, ThiE, ThiL, and a sulfur source. Loss of any abolishes de novo ThDP. - *Conditional / interchangeable:* the dehydroglycine-generating step (ThiO vs ThiH) and the sulfur-relay architecture (ThiI ± rhodanese domain; alternative desulfurases) are lineage- and oxygen-dependent (P 22773787 P 21724998).
- *Accessory / shared:* the **TenI** thiazole tautomerase optimizes but is not strictly required for coupling; Dxs and IscS also serve other pathways; salvage enzymes and transporters (§2) support ThDP pools but are not de novo steps.

4. Major molecular players and active assemblies

- **ThiC — phosphomethylpyrimidine synthase.** Homodimer; N-terminal novel domain + central (βα)8 barrel + C-terminal [4Fe-4S]-binding CX2CX4C. Radical-SAM superfamily; evolutionarily related to adenosylcobalamin radical enzymes (P 18953358). Oxygen-sensitive.
- **ThiD (ThiD/J) — HMP/HMP-P kinase.** Bifunctional; ribokinase-like fold; delivers HMP-PP (P 10075431).
- **Dxs — DXP synthase.** ThDP-dependent transketolase-family enzyme; branch point with the MEP isoprenoid pathway; rate-limiting for MEP flux and an antibacterial target (P 41994131 P 40845551).
- **ThiO — glycine oxidase.** FAD-dependent; homotetramer in *B. subtilis* (structurally akin to D-amino-acid/sarcosine oxidase; hydride-transfer mechanism) (P 12627963). Species-variable oligomer/substrate preference — *P. putida* ThiO is monomeric and glycine-preferring, whereas *Bacillus*/*Geobacillus* prefer D-proline (P 26875494).
- **ThiH — tyrosine lyase.** Radical-SAM [4Fe-4S] enzyme; forms a ThiGH heterodimer in *E. coli*; converts tyrosine to dehydroglycine + p-cresol (P 19923213).
- **IscS — cysteine desulfurase.** PLP-dependent homodimer; persulfide on a conserved Cys; central sulfur hub (P 12382038).
- **ThiF / ThiS — ubiquitin-like sulfur-carrier system.** ThiF (E1-like adenylyltransferase) activates the C-terminal Gly of ThiS (P 9632726).

The 2.0 Å ThiS–ThiF co-structure shows ThiS structurally resembles **ubiquitin and MoaD** and ThiF resembles the **ubiquitin-activating E1 and MoeB**; sulfur transfer proceeds via a ThiS–ThiI acyl-disulfide and a ThiF-Cys184 disulfide-interchange (P 16388576).

- **TenI — thiazole tautomerase (accessory).** A ThiE-paralog fold that isomerizes the ThiG product to the correct tautomer for coupling; not universally essential (P 21534620).
- **ThiI — sulfurtransferase.** In Gammaproteobacteria, an adenyltransferase + C-terminal rhodanese domain (Cys persulfide relay); in Firmicutes, lacks the rhodanese domain and partners a distinct desulfurase (NifZ) (P 22773787 P 21724998 P 10722656).
- **ThiG — thiazole synthase.** (βα)8 barrel; convergent condensation node (P 15489164).
- **ThiE — thiamine-phosphate synthase.** (βα)8 barrel; couples the two moieties (P 22226942). Not universal at the sequence level: most archaea instead use a **ThiN** domain fused to ThiD that catalyzes the same reaction but shares no sequence similarity with ThiE (an analog, not an ortholog) (P 24583237); eukaryotes fuse the coupling activity with a salvage HET-kinase in bifunctional **THI6** (P 20968298).
- **ThiL — thiamine-monophosphate kinase.** ATP-grasp/AIRS-like fold; in-line phosphoryl transfer with metal (Mg2+) coordination; validated drug target (P 30867460

P 32404369

P 42103215
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Regulation. Thiamine genes are governed by **ThDP riboswitches** that sense the end product and repress transcription/translation of thi operons (e.g., thiC, thiM), providing negative feedback
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P 26932506
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The pyrithiamine sensitivity of these riboswitches makes them themselves antibacterial targets
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P 21234302
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5. Evolutionary and cell-biological variation

- **Deep, mosaic ancestry.** The pathway assembles several ancient protein families independently recruited to thiamine: the **radical-SAM superfamily** (ThiC, ThiH), **PLP desulfurases** (IscS), **ThDP-transketolase** enzymes (Dxs), and the **ubiquitin-like β-grasp sulfur carriers** (ThiS/ThiF). The 2.0 Å ThiS–ThiF structure shows ThiS resembles ubiquitin/MoaD and ThiF resembles E1/MoeB
(
P 16388576
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and archaeal noncanonical E1 enzymes (UbaA) with ubiquitin-like SAMPs use the same E1/MoeB/ThiF superfamily chemistry for both molybdopterin/tRNA sulfur transfer and protein conjugation
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P 21368171

P 27459543
).
ThiS/ThiF chemistry is thus a molecular fossil of pre-ubiquitin sulfur biochemistry, and ThiS/MoaD/Urm1 are the best representatives of the ancestral sulfur-carrier role.
- **Aerobic vs anaerobic wiring of dehydroglycine.** ThiO (flavin, O2-using) is favored by aerobes; ThiH (radical-SAM, O2-sensitive) by facultative/anaerobes. Both converge on the same dehydroglycine substrate for ThiG — a striking example of non-homologous enzymes solving the same supply problem under different redox regimes

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P 12627963
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P 19923213

- **Sulfur-relay divergence.** Gammaproteobacterial ThiI carries a rhodanese domain and uses IscS; most Firmicutes ThiI lack this domain and use NifZ, and additional ThiI/IscS-independent, cysteine-dependent routes to thiazole exist
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P 22773787
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P 21724998
Archaea can incorporate sulfide directly.
- **The coupling enzyme is not universal.** Pyrimidine–thiazole condensation is a fixed chemical requirement met by at least three different protein solutions: standalone **ThiE** in most bacteria; a **ThiN** domain fused onto ThiD in most archaea, which is functionally analogous but non-homologous to ThiE
(
P 24583237
);
and a bifunctional **THI6** fusion (coupling + salvage HET-kinase) in eukaryotes
(
P 20968298
).
Sequence orthology of ThiE therefore under-reports pathway presence.
- **Final-step divergence from eukaryotes.** Bacteria typically make ThDP directly by ThiL phosphorylation of TMP; eukaryotes and some bacteria instead dephosphorylate TMP to free thiamine and then pyrophosphorylate it via thiamine pyrophosphokinase (TPK/ThiN)
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P 26639844
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P 16291685
This is the sharpest bacterial/eukaryotic contrast and underlies ThiL's appeal as a selective target.
- **Thiazole synthesis is not universal.** The bacterial ThiG/ThiS route is entirely distinct from the eukaryotic **THI4** route, in which a single "suicide" enzyme uses an active-site Cys as sacrificial sulfur donor for one catalytic cycle and is then degraded — an energetically costly design
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P 30337452

P 36920242).

Some prokaryotic THI4s are non-suicidal, sulfide-dependent enzymes, informing the ancestral reconstruction and bioengineering of low-cost thiazole synthesis.

- **Genome reduction and auxotrophy.** Streamlined and host-associated bacteria frequently lose de novo steps and rely on transport/salvage; e.g., some anaerobes and *Dehalococcoides* require thiamine as a growth supplement (P 23479751).

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6. Constraints, dependencies, and failure modes

Obligatory ordering. - HMP-P (ThiC) must precede HMP-PP (ThiD): only the diphosphate is a ThiE substrate. - All three thiazole inputs (DXP via Dxs; dehydroglycine via ThiO/ThiH; sulfur via the IscS→ThiI/ThiF→ThiS relay) must be available for ThiG. - Within the sulfur relay the order is fixed: IscS persulfide → ThiF adenylation of ThiS → ThiS-thiocarboxylate → ThiG. - Both branches must complete before ThiE coupling; ThiE must precede ThiL.

Mutually exclusive / condition-specific choices. - ThiO (aerobic, flavin) vs ThiH (anaerobic, radical-SAM) are alternative, essentially mutually exclusive solutions selected by oxygen availability and lineage. - The oxygen-sensitivity of the [4Fe-4S] enzymes ThiC and ThiH constrains when/where those steps operate and complicates in vitro reconstitution.

Evidence that rules out otherwise plausible paths. - The thiamine pyrimidine does **not** come from pyrimidine-nucleotide metabolism: ThiC's substrate is the purine intermediate AIR (P 18953358). - Sulfur is delivered as a **protein-bound thiocarboxylate**, not free sulfide: ThiS-thiocarboxylate is the immediate donor, and 18O labeling shows ring oxygen comes from DXP, not buffer (P 15489164 P 9632726).

- ThiL, not TPK, performs the final step in most bacteria — de novo bacterial ThDP does not transit free thiamine (P 26639844).

Failure modes / vulnerabilities. - Loss of any obligatory enzyme yields thiamine auxotrophy (or bradytrophy where salvage compensates, e.g., *B. subtilis* ΔthiL; P 16291685).
- ThiL is essential for both biosynthesis and salvage and its inhibition/destabilization is bactericidal in pathogens (*Acinetobacter*, *Pseudomonas*) (P 32404369 P 42103215).
- Dxs and ThiE are proposed broad-spectrum targets; the ThDP riboswitch is targeted by pyrithiamine (P 21234302).

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7. Controversies and open questions

- **ThiC mechanism.** The radical-SAM rearrangement of AIR→HMP-P is chemically extraordinary and the complete sequence of radical intermediates is still not fully resolved; the enzyme's oxygen sensitivity limits mechanistic study (P 18953358).
- **ThiI's exact role in thiazole synthesis.** Genetic data show the rhodanese domain is sufficient in *Salmonella*, yet residues needed for 4-thiouridine differ from those for thiazole, and ThiI/IscS-independent sulfur routes exist — implying current annotations over-generalize a single mechanism (P 21724998 P 22773787). How sulfur reaches thiazole in Firmicutes and archaea remains partly open.

- **Organism-mixing in the literature.** Much mechanistic detail is stitched together from *E. coli*, *B. subtilis*, *Salmonella*, and *Pseudomonas*, which differ in the ThiO/ThiH choice, ThiI architecture, and salvage wiring. Claims should be qualified by organism; the "canonical" pathway is a composite.
- **Sulfur-donor identity in vivo.** Whether cysteine-derived persulfide, free sulfide, or alternative carriers dominate varies with organism and redox state; redundant routes complicate clean genetic assignment (P 21724998).
- **Evolutionary origin of the ThiS/ubiquitin connection.** ThiS/ThiF strongly resemble ubiquitin/MoaD-MoeB; whether thiamine sulfur transfer or molybdopterin synthesis is the more ancestral use of this β-grasp chemistry is unsettled.
- **Engineering low-cost thiazole synthesis.** The energetic cost of eukaryotic suicide THI4 versus catalytic bacterial ThiG has driven efforts to install non-suicidal / sulfide-dependent thiazole synthases, but these are slow and O2-sensitive — an open bioengineering problem (P 30337452 P 36920242).

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Explicit statements of uncertainty

- The "canonical" pathway described here is a **composite** across model organisms; the ThiO/ThiH choice, ThiI architecture, salvage wiring, and final activation step differ by lineage and should not be generalized to all bacteria.
- Several claims (radical intermediates of ThiC; in vivo sulfur-donor identity; ThiI's precise catalytic role in thiazole) rest on partial or indirect evidence.
- This synthesis is literature-based; no primary experimental data were generated here.
